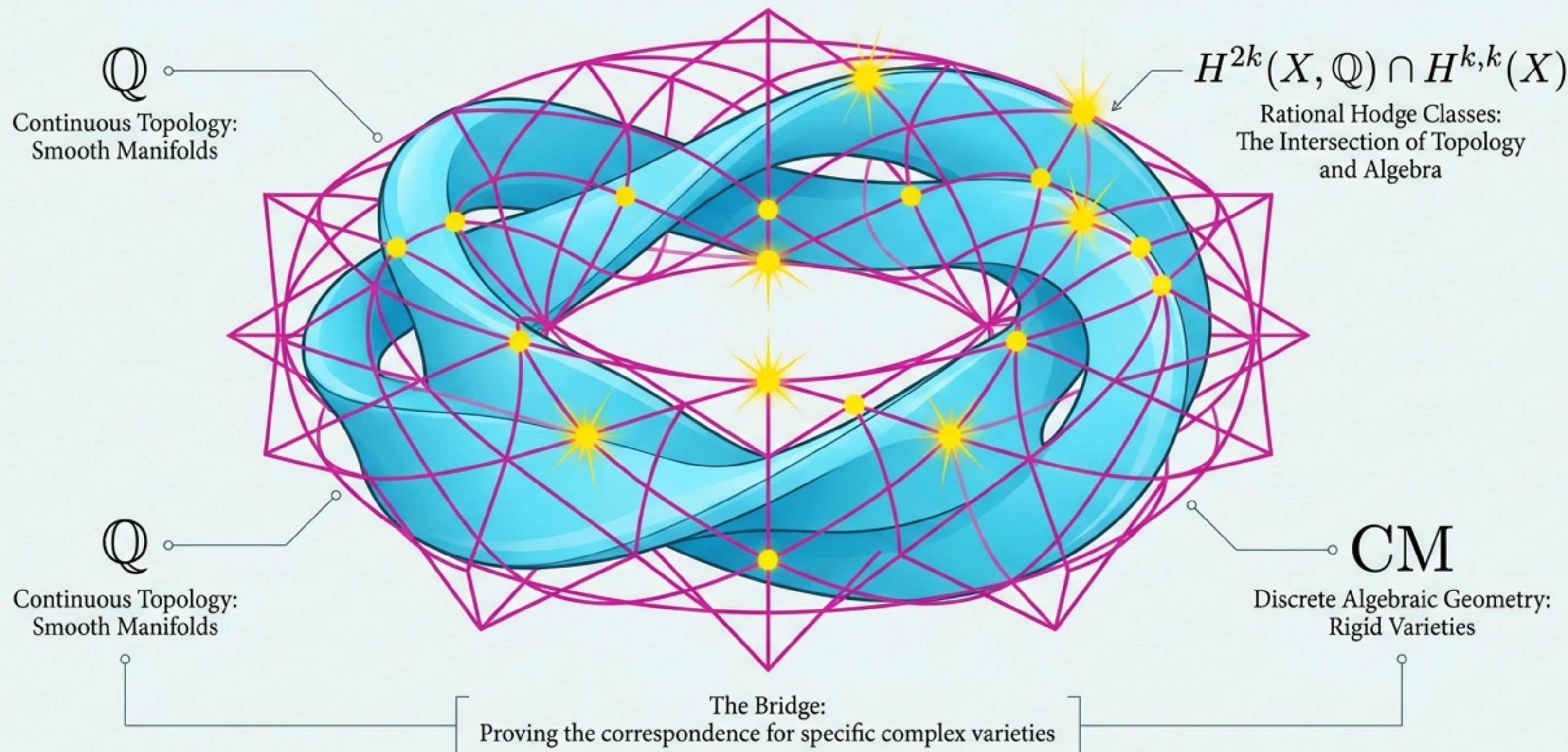


THE RATIONAL HODGE CONJECTURE FOR CM ABELIAN VARIETIES

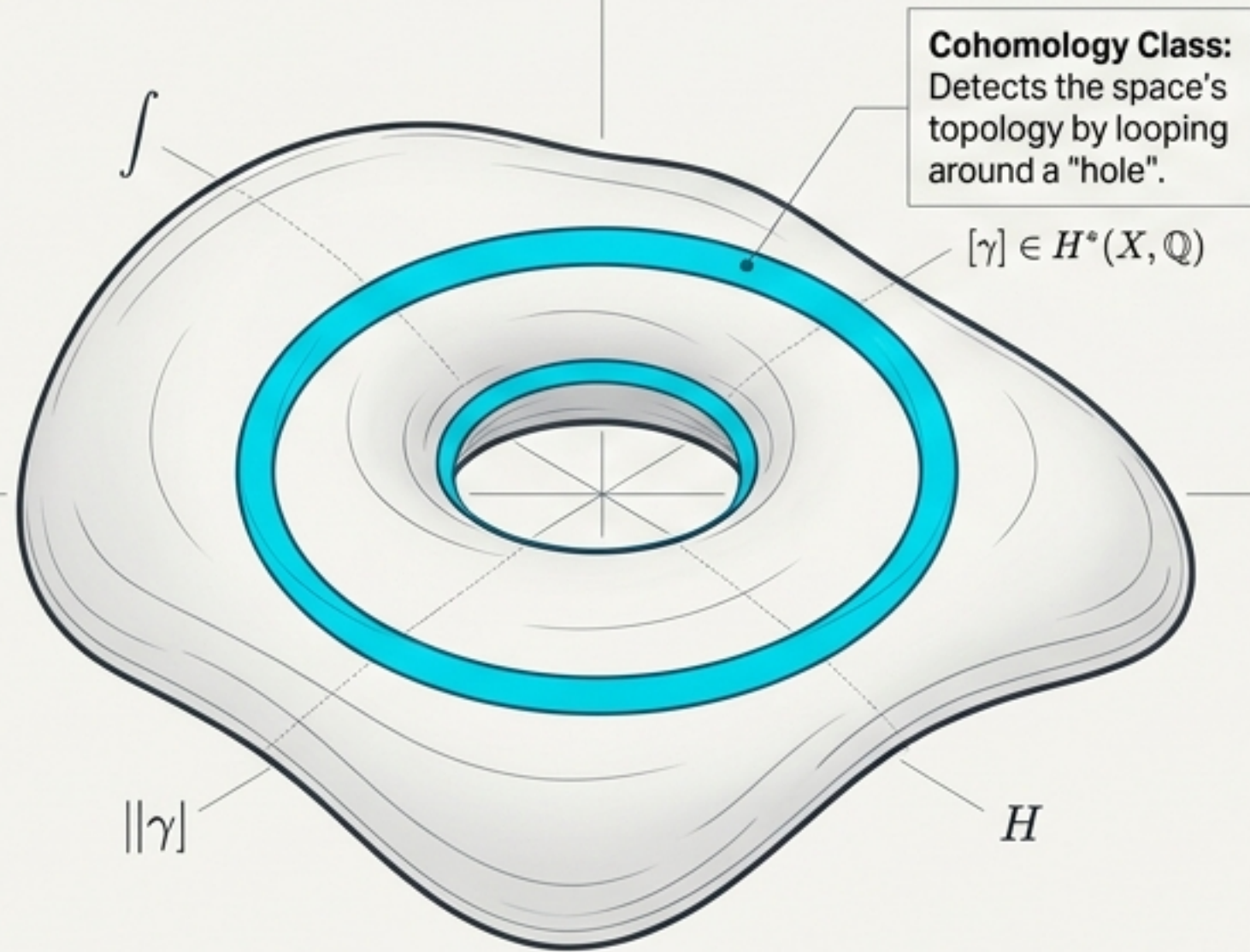
Demystifying the September 2026 OpenAI Preprint: A visual guide bridging topology and algebraic geometry.



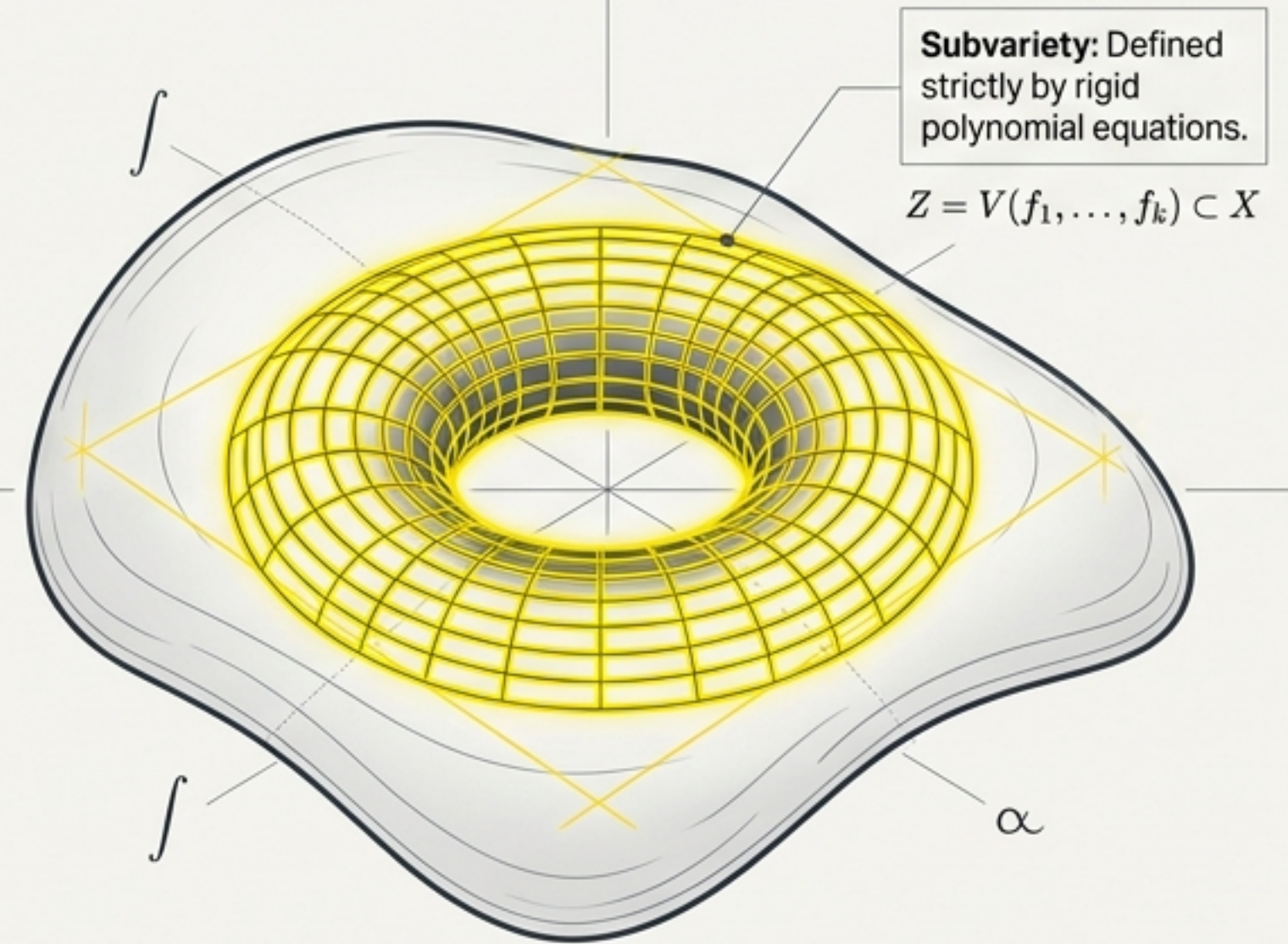
A Millennium Problem: Topology vs. Algebraic Geometry

Formulated by W.V.D. Hodge in 1950 and named a Clay Millennium Problem in 2000, the Hodge Conjecture asks a profound question about the shape of higher-dimensional spaces: Can every specific type of topological hole perfectly align with a combination of algebraic subvarieties?

Topology



Algebraic Geometry



The Mathematical Arena: Complex Projective Varieties

The Hodge Conjecture lives entirely within highly structured mathematical spaces. To decode it, we must map how subvarieties generate cohomology.

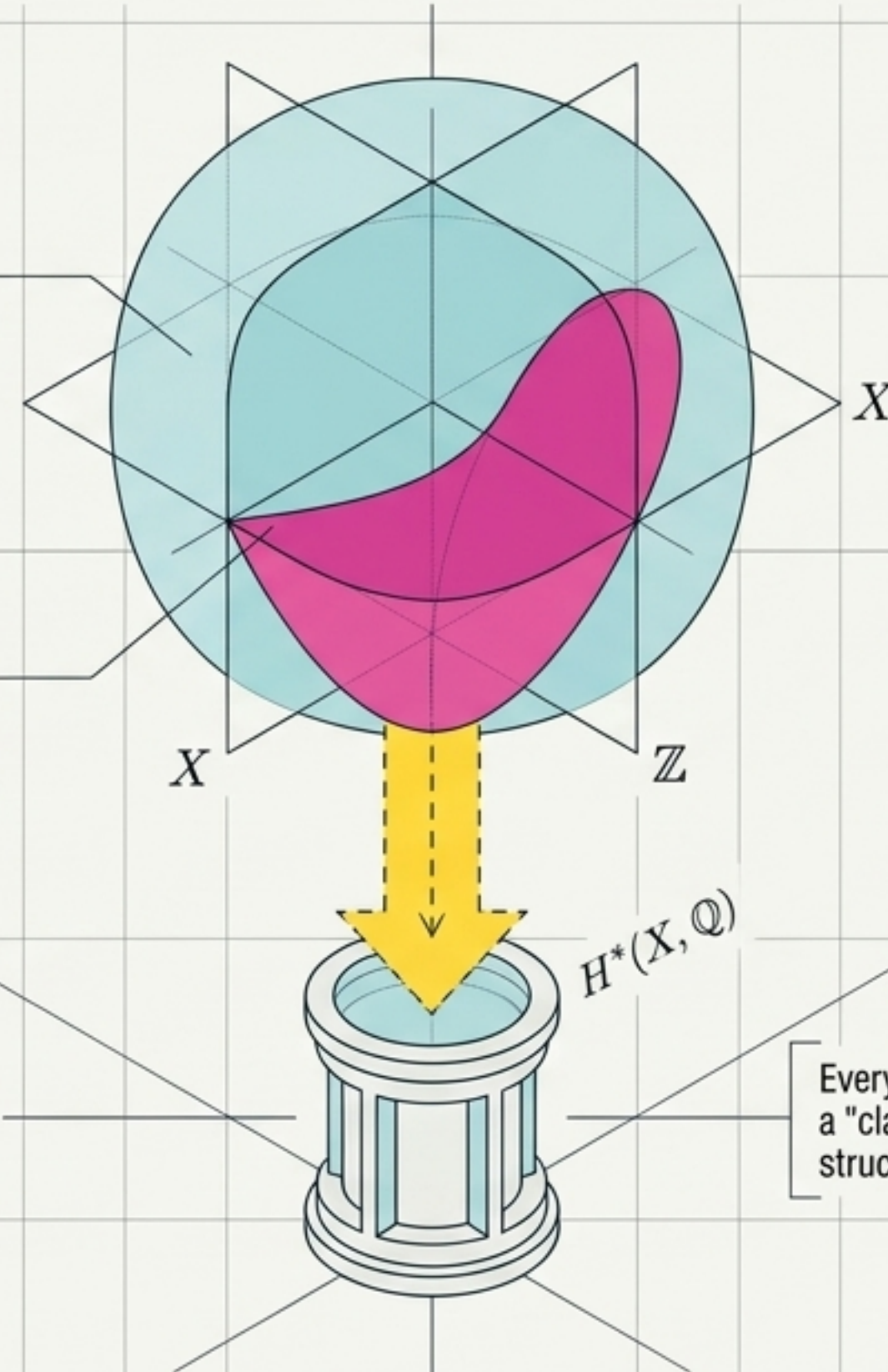
Main Variety

A shape defined by polynomial equations in projective space.

Subvariety

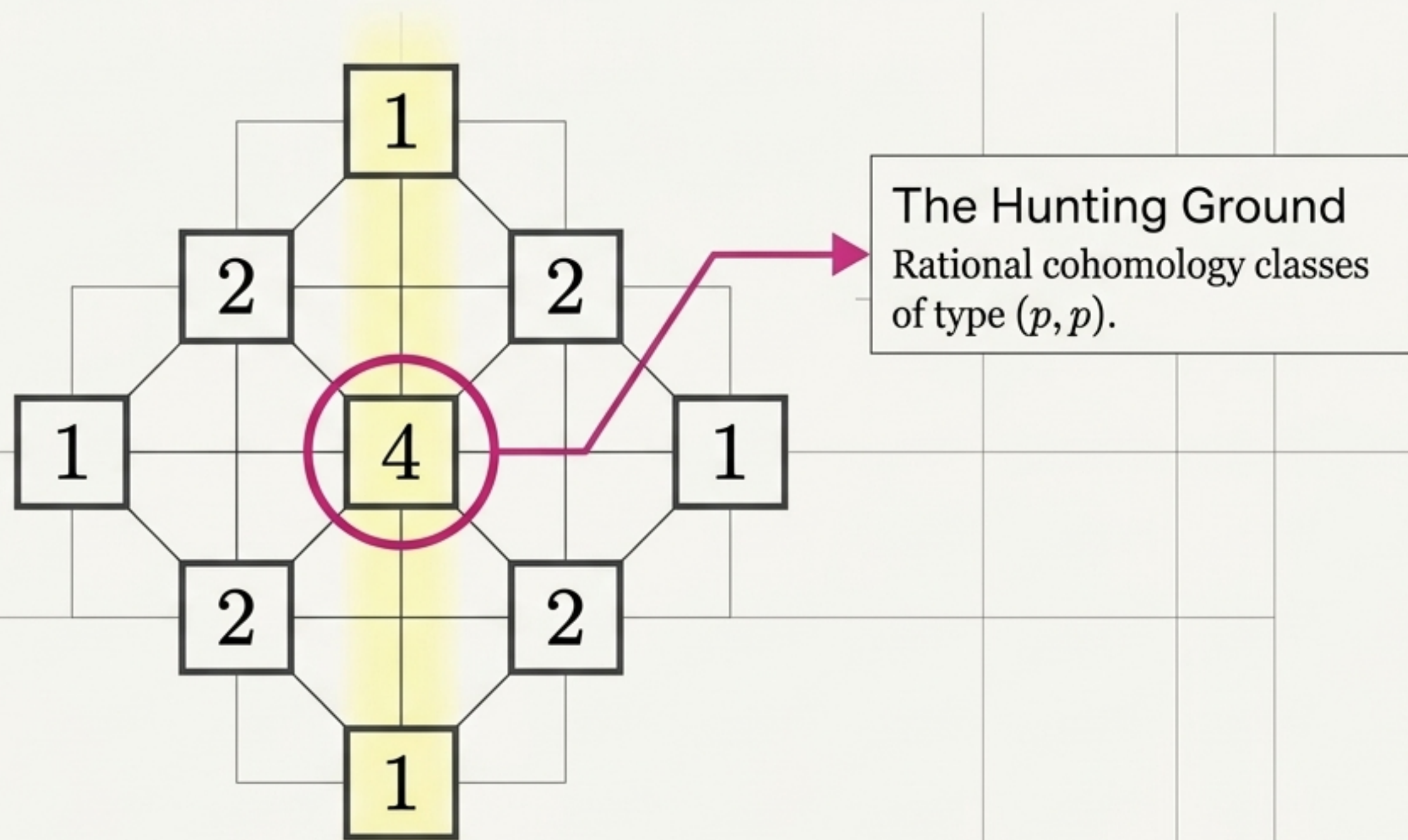
A smaller polynomial shape living within the main space.

Cohomology Group



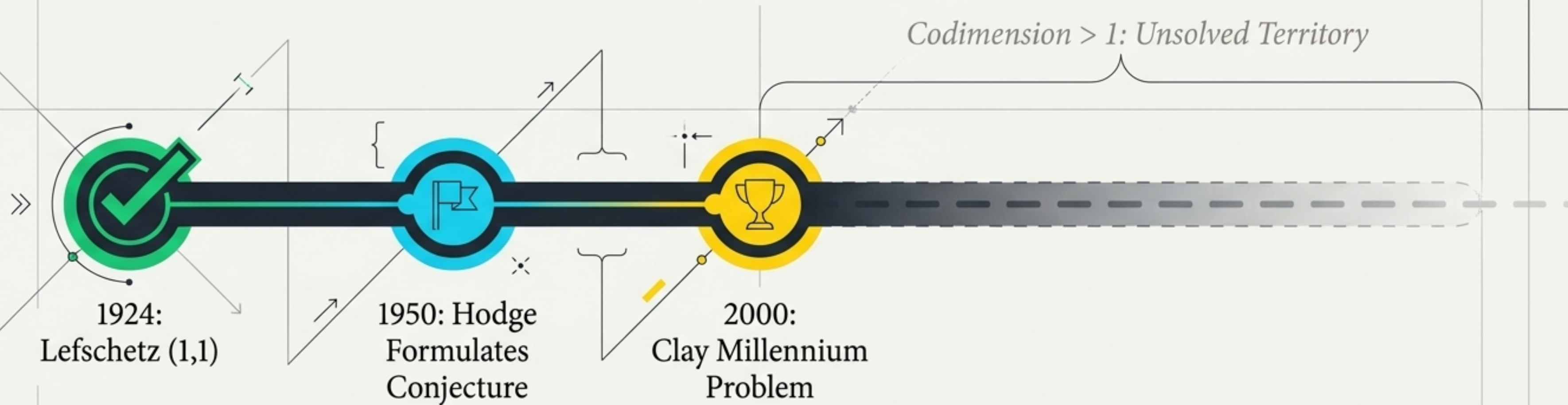
Slicing Cohomology: The Hodge Diamond

On a complex manifold, cohomology fractures into (p, q) types based on differential forms (p = holomorphic, q = anti-holomorphic). The Hodge Conjecture searches exclusively within the perfectly symmetric (p, p) center.



The Benchmark: The Lefschetz (1,1) Theorem

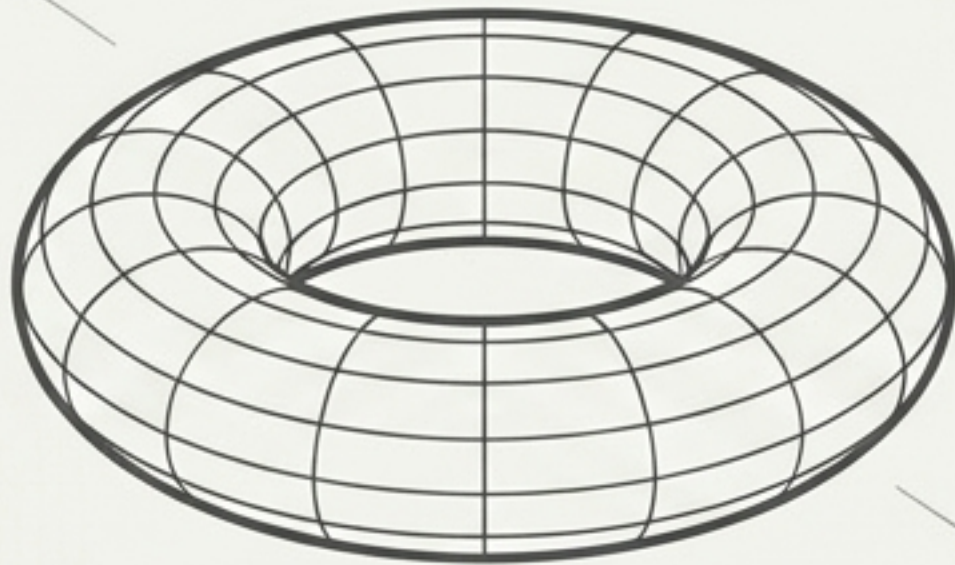
The simplest case of the Hodge Conjecture was solved before the conjecture was fully formulated. The 1924 Lefschetz (1,1) theorem proved that in codimension 1, every rational Hodge class comes from a combination of divisors. Higher dimensions remained a total mystery.



The Chosen Spaces: CM Abelian Varieties

The September 2026 preprint does not solve the conjecture for all shapes. It focuses on a highly symmetric class of spaces known as CM Abelian Varieties—projective complex tori with a precise group structure.

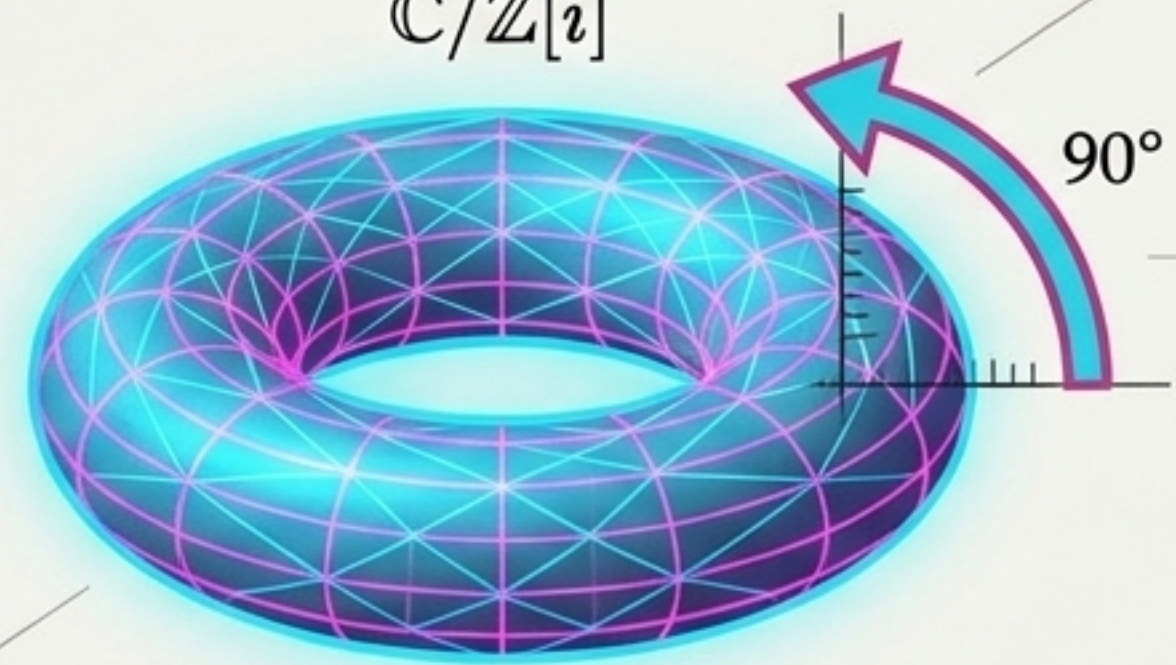
Standard Elliptic Curve



Normal Symmetries

CM Elliptic Curve

$$\mathbb{C}/\mathbb{Z}[i]$$



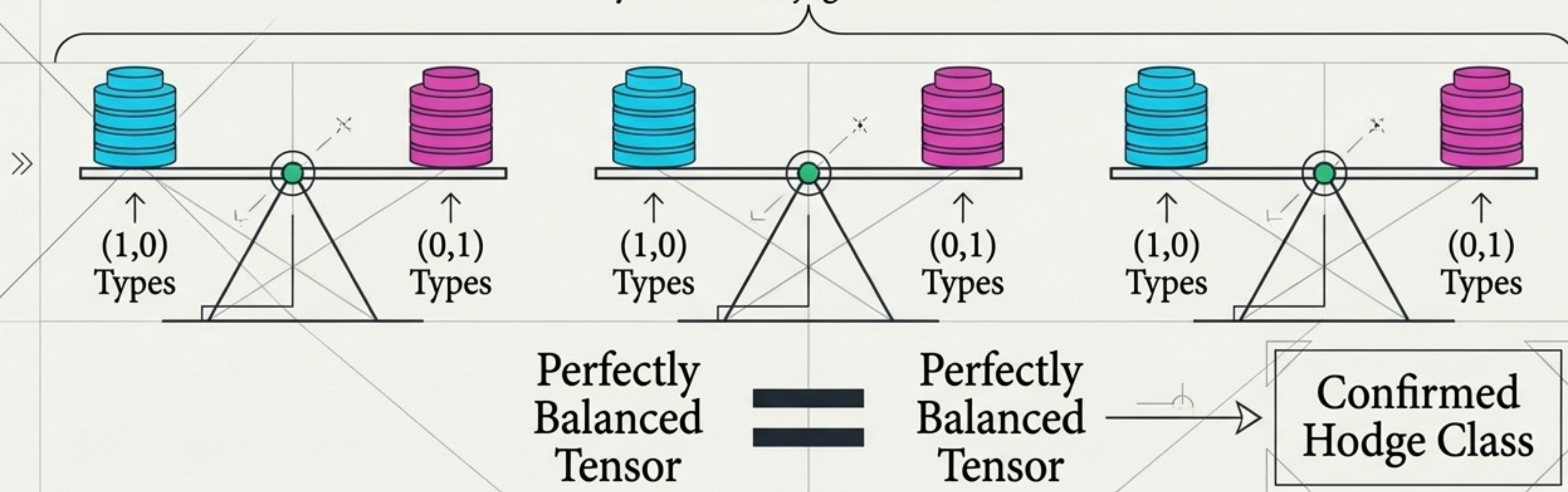
Complex Multiplication (CM)

Possesses extra hidden symmetries where $\text{End}(A) \otimes \mathbb{Q}$ contains a commutative semisimple algebra of dimension $2 \cdot \dim A$. The lattice perfectly rotates into itself.

Pohlmann's Criterion: The Balance of Symmetries

How do we identify a Hodge class on a CM Abelian variety? In 1968, Pohlmann established that the first cohomology group perfectly splits into embedding lines of type $(1,0)$ or $(0,1)$. A tensor of these lines is a Hodge class exactly when perfectly balanced across every single Galois conjugate.

Every Galois Conjugate must balance.



The Breakthrough: Theorem 1.1 (September 2026)

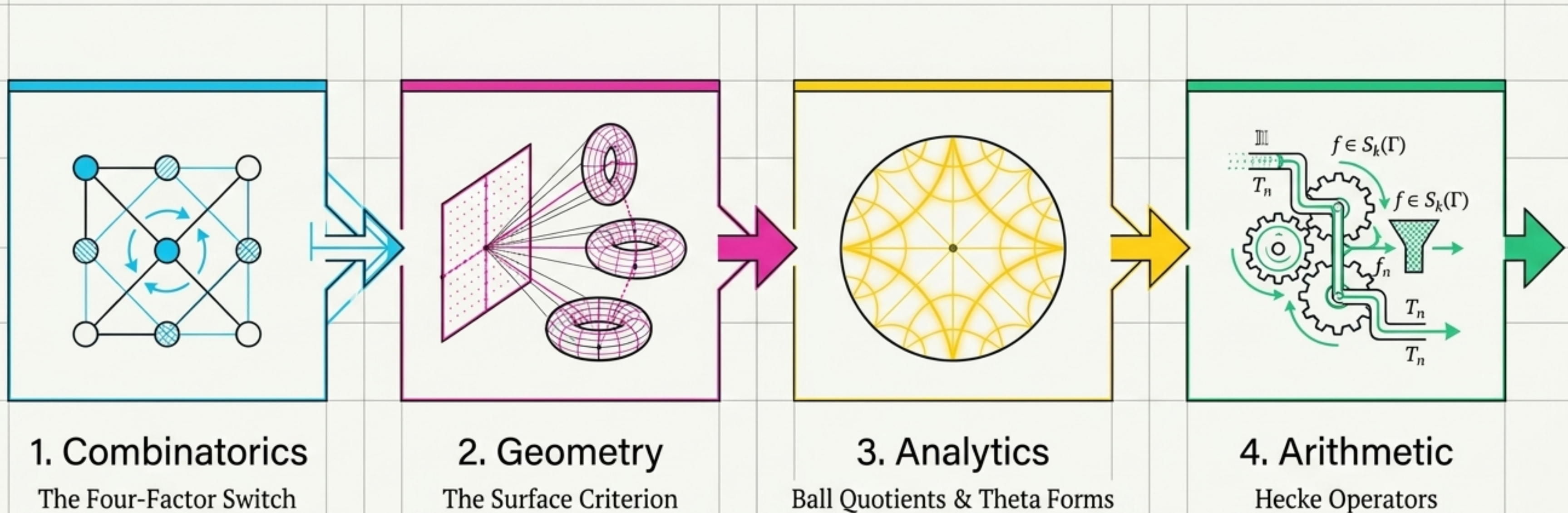
The core claim of the unrefereed OpenAI preprint is a massive leap forward for algebraic geometry, settling the conjecture for an entire universe of symmetric spaces.

On every complex CM abelian variety, and on all their products and powers, every rational Hodge class in every codimension is algebraic.

Theorem 1.1, unrefereed preprint

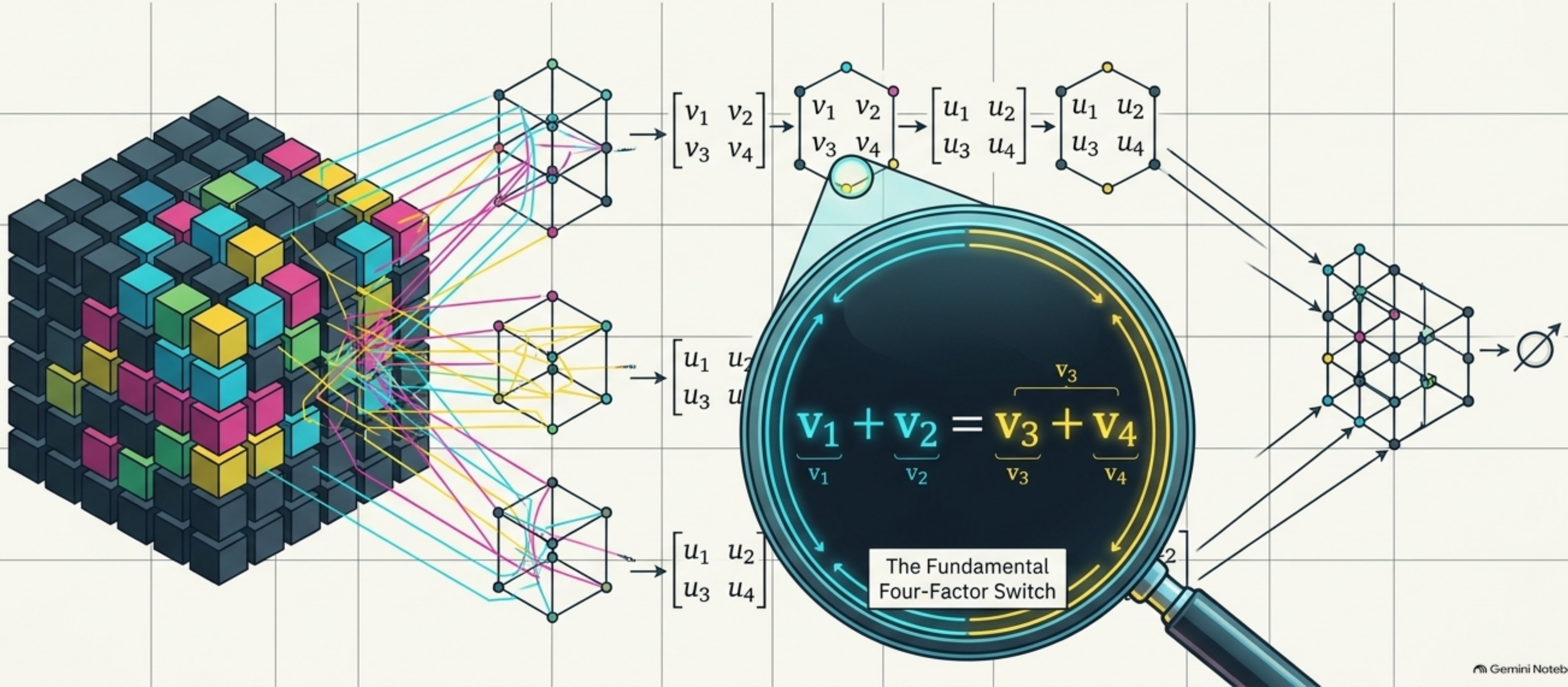
The Engine: A Four-Stage Mathematical Pipeline

Proving Theorem 1.1 required weaving together four vastly different fields of mathematics. The authors built an interconnected pipeline to force the Hodge classes to be algebraic.



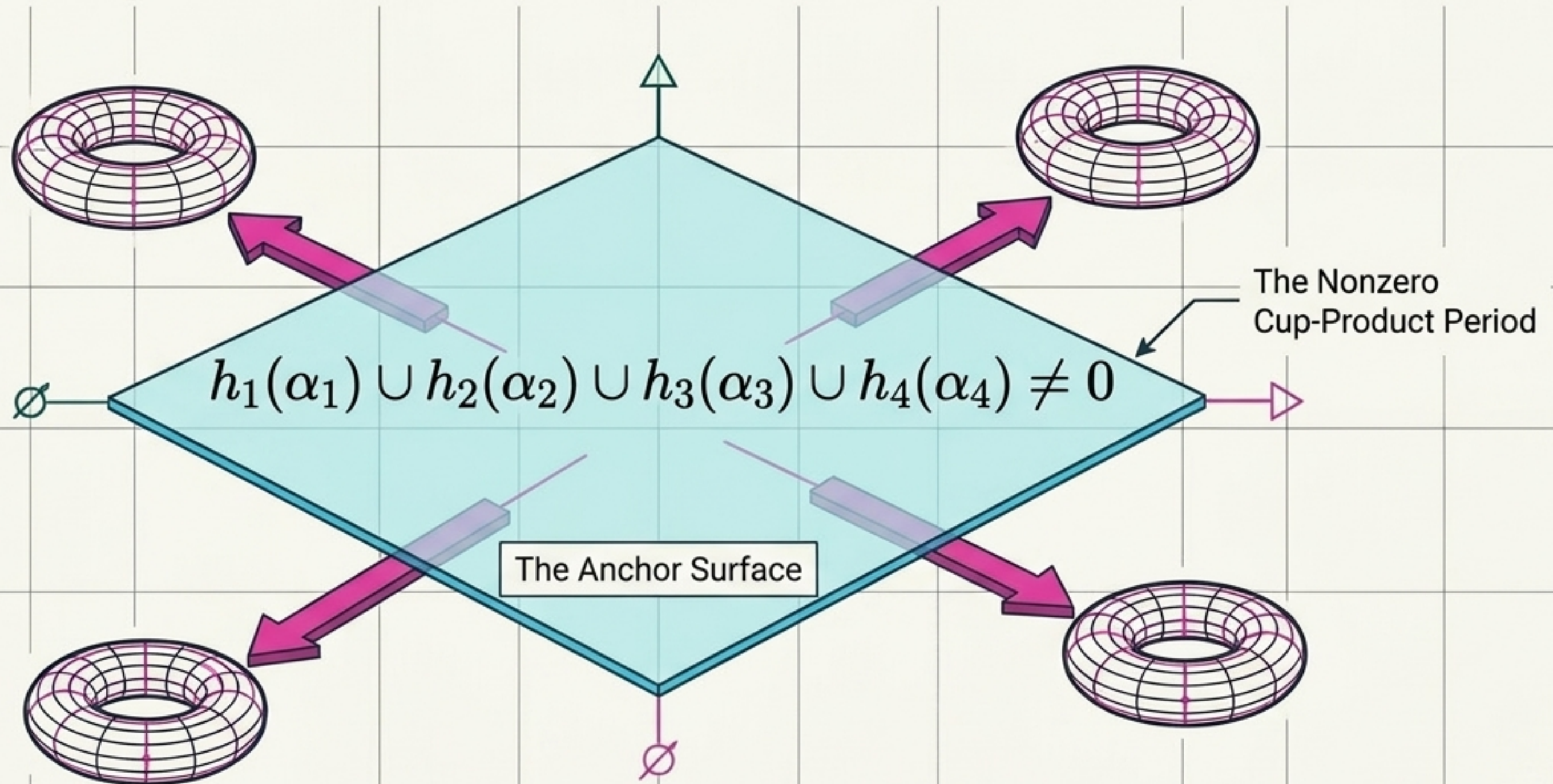
Stage 1: Combinatorics and the Four-Factor Switch

The proof begins by dismantling complex balanced tensors. Any balanced list of CM types can be linked by simple two-row swaps. If this base switch is proven algebraic, the entire chain becomes algebraic.



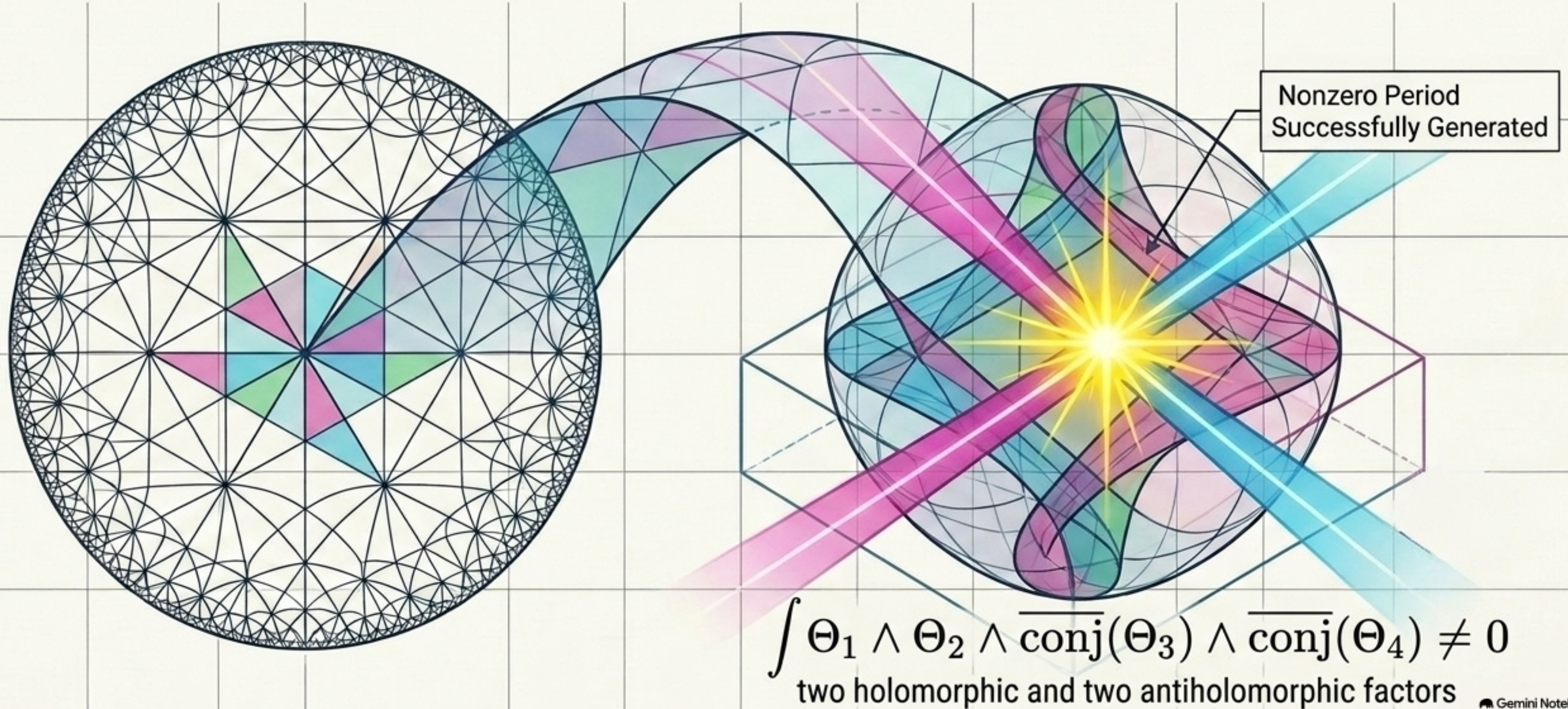
Stage 2: Geometry and the Surface Criterion

To prove the 4-factor switch is algebraic, you anchor it to geometry. Finding just one smooth surface that maps to four CM abelian varieties—and yields a nonzero cup-product period—is enough.



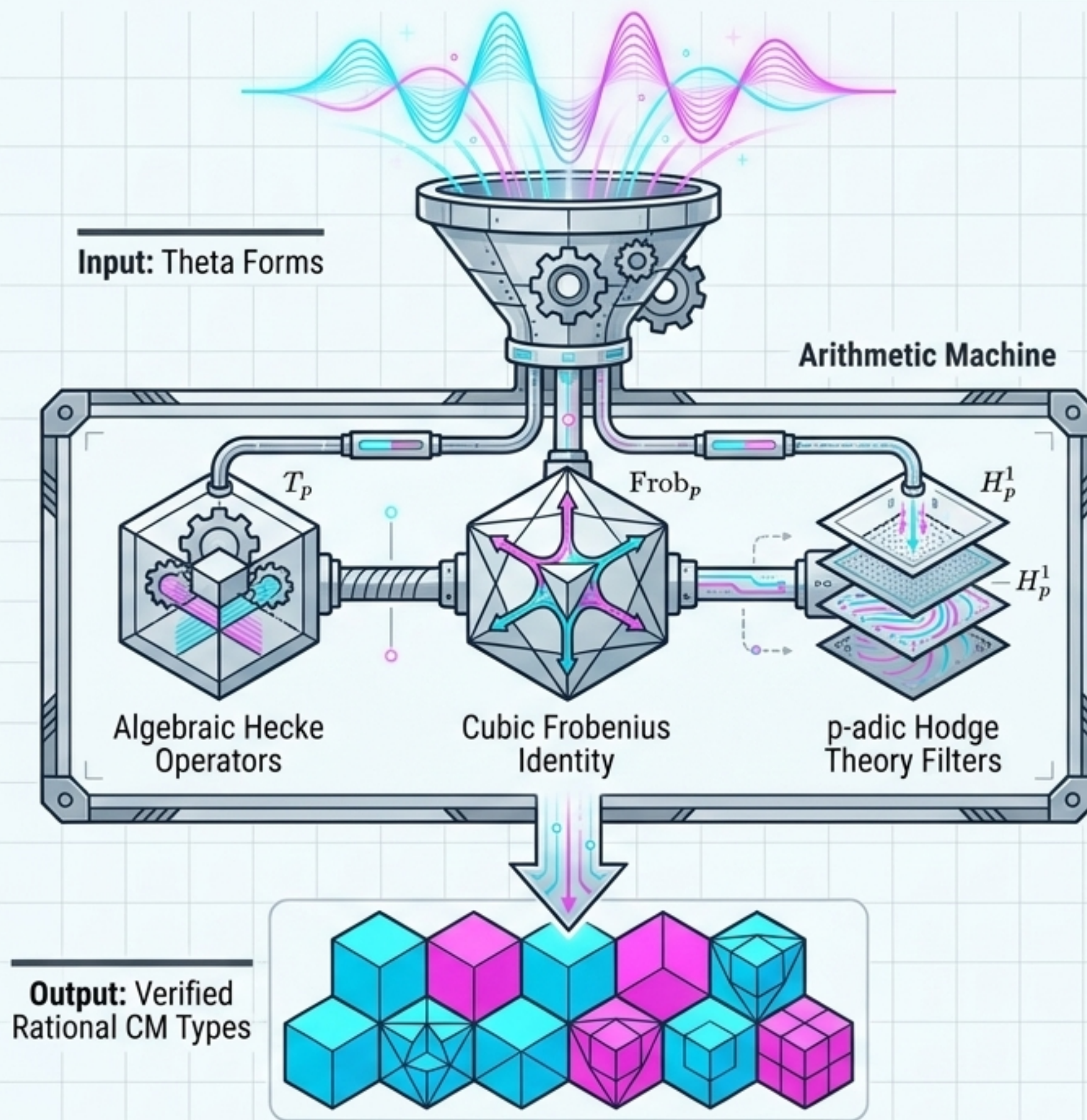
Stage 3: Analytics, Ball Quotients, and Theta Forms

To construct this elusive surface, the proof moves into complex analytics. By utilizing the Weil representation on a complex 2-ball, specific holomorphic 'theta one-forms' are wedged together to generate the required period.



Stage 4: Arithmetic Sourcing

The final step ensures these analytic forms originate from the correct rational CM types. Algebraic operators and p-adic theory lock down the specific slopes to confirm the exact CM sources.



The Domino Effect: Ripples Through Mathematics

Combined with theorems of J. S. Milne (1999, 2002) and his later addenda, Theorem 1.1 gives two further results for abelian varieties.



The Uncharted Territory: What Remains Open?

While Theorem 1.1 is a monumental leap, the war is not over. The Hodge conjecture for general varieties remains unproven, and this preprint represents the bleeding edge of current research.

